

Unit 3.1 Data Handling and Analysis

Paper 1: Basic

1. 25 32 16 19 40 32 14 32

(a) Find the median of the set of numbers.

[1]

(b) Find the range of the set of numbers.

[1]

[N22/I/3]

2. Five positive integers have a mean of 12, a median of 15 and a mode of 21.
Find the five numbers.

[2]

[N21/I/6]

3. The table shows how a group of students travel to school.

	Frequency	Sector angle
Walk	24	
Bus	21	
Cycle	9	

A pie chart is to be drawn to show this information.

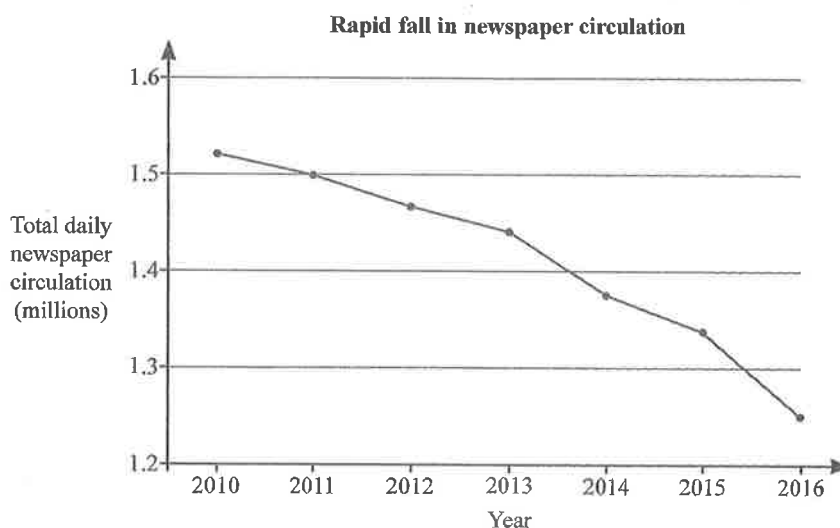
Calculate the angle of each sector of the pie chart.

Write your answers in the table.

[3]

[N21/I/10]

4. The graph shows the total daily newspaper circulation at the end of each of the given years.



(a) State one misleading feature of the graph.

[1]

(b) Explain how this feature affects the reader's interpretation of the graph.

[1]

[N20/I/4]

5. The time, to the nearest minute, that each of 23 men spent shopping one week was recorded. The results are shown on the stem-and-leaf diagram.

7	5	6	8	9	9	9		
8	0	0	1	1	2	3	4	7
9	3	4	4	5	6	8		
10	0	1	2					

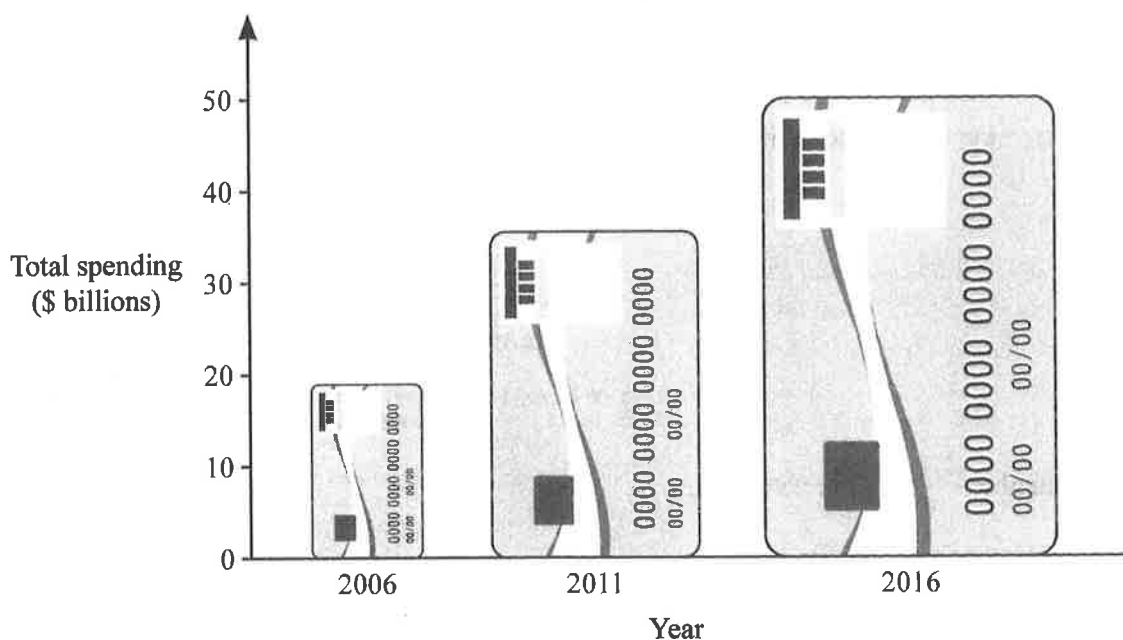
Key: 7 | 5 represents 75 minutes

- (a) Write down the median of the times. [1]
- (b) When the time for a 24th man is added to the diagram, the range increases to 33 minutes. Find the two possible times that this man spent shopping. [1]

[N20/I/7]

6.

Total credit card and charge card spending in Singapore



Explain how the chart above may be misleading.

[1]
[N19/I/2]

7. The stem-and-leaf diagram shows the masses, in grams, of some apples.

19	1	4	5	5			
20	0	3	4	7	8	8	
21	0	1	4	7	7	8	9
22	4	6	8				

Key: 19 | 4 represents 194 grams

For these masses, find

- (a) the range, [1]
- (b) the median. [1]

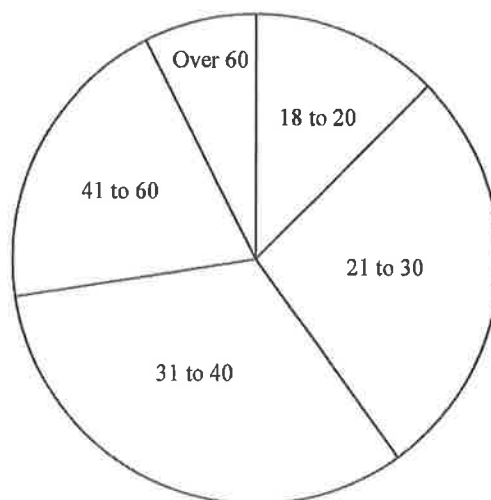
[N17/I/3]

8. A group of 200 adults took part in a challenge.

The table below shows the distribution of the times taken to complete the challenge.

Time (t minutes)	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 50$	$50 < t \leq 60$	$60 < t \leq 70$
Number of adults	28	84	62	21	5

- (a) Calculate an estimate of the mean time. [1]
- (b) Calculate an estimate of the standard deviation of these times. [1]
- (c) This **accurate** pie chart shows the age groups of a second group of adults taking part in the challenge.



44 of the adults were aged 21 to 30 years old.

Find the number of adults that were aged 41 to 60 years old.

[2]
[N17/I/16]

Paper 1: Intermediate

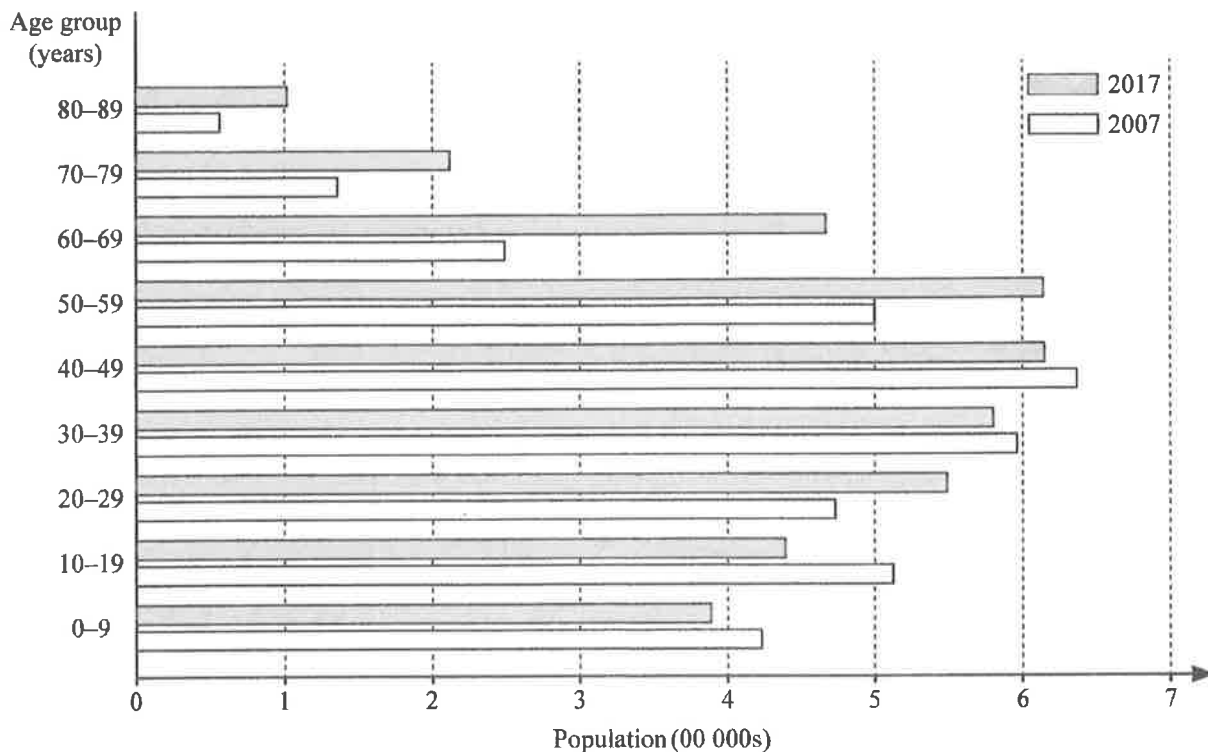
9. The table shows the monthly salaries of a group of 30 employees.

Monthly salary, m (\$)	Frequency
$2400 \leq m < 2500$	3
$2500 \leq m < 2600$	5
$2600 \leq m < 2700$	14
$2700 \leq m < 2800$	8

- (a) Calculate an estimate for
- (i) the mean monthly salary of the employees, [1]
- (ii) the standard deviation of the monthly salaries. [1]
- (b) Each of the employees is to be given a salary increase of \$60 per month.
- Explain how the mean and standard deviation will change after the salary increase. [1]

[N22/I/24]

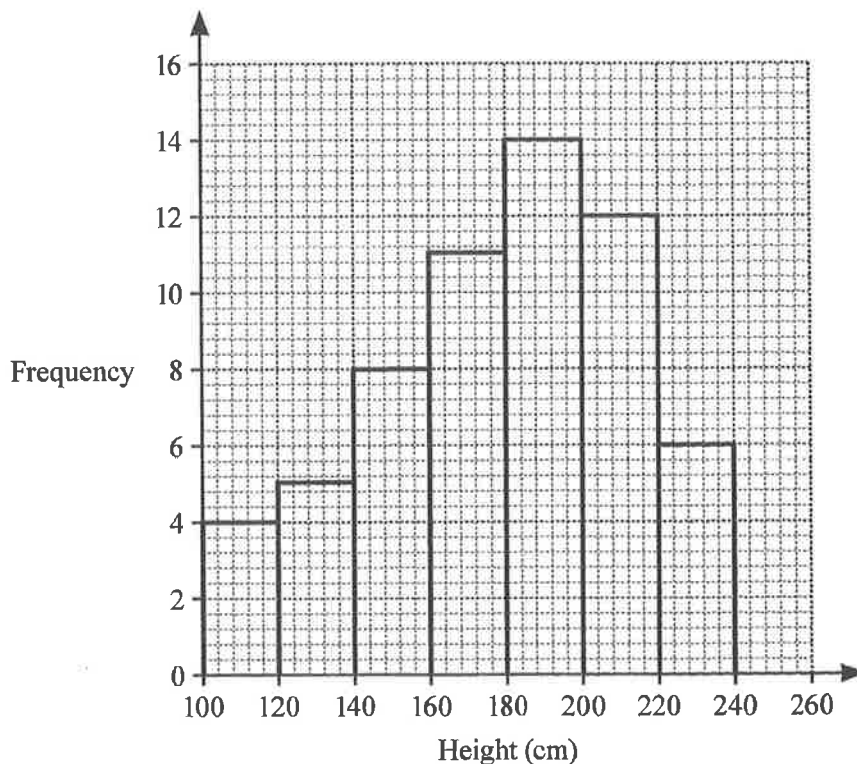
10. The graph shows information about the ages of the resident population of Singapore in 2007 and 2017.



- (a) Estimate the number of people in 2017 that were aged 60–69. [1]
- (b) What feature of the graph supports the fact that the birth rate has fallen in Singapore between 2007 and 2017? [1]
- (c) Explain what the graph shows about the change in the mean age of the resident population of Singapore in 2007 and 2017. [2]
- Justify your answer with references to the graph.

[N21/I/13]

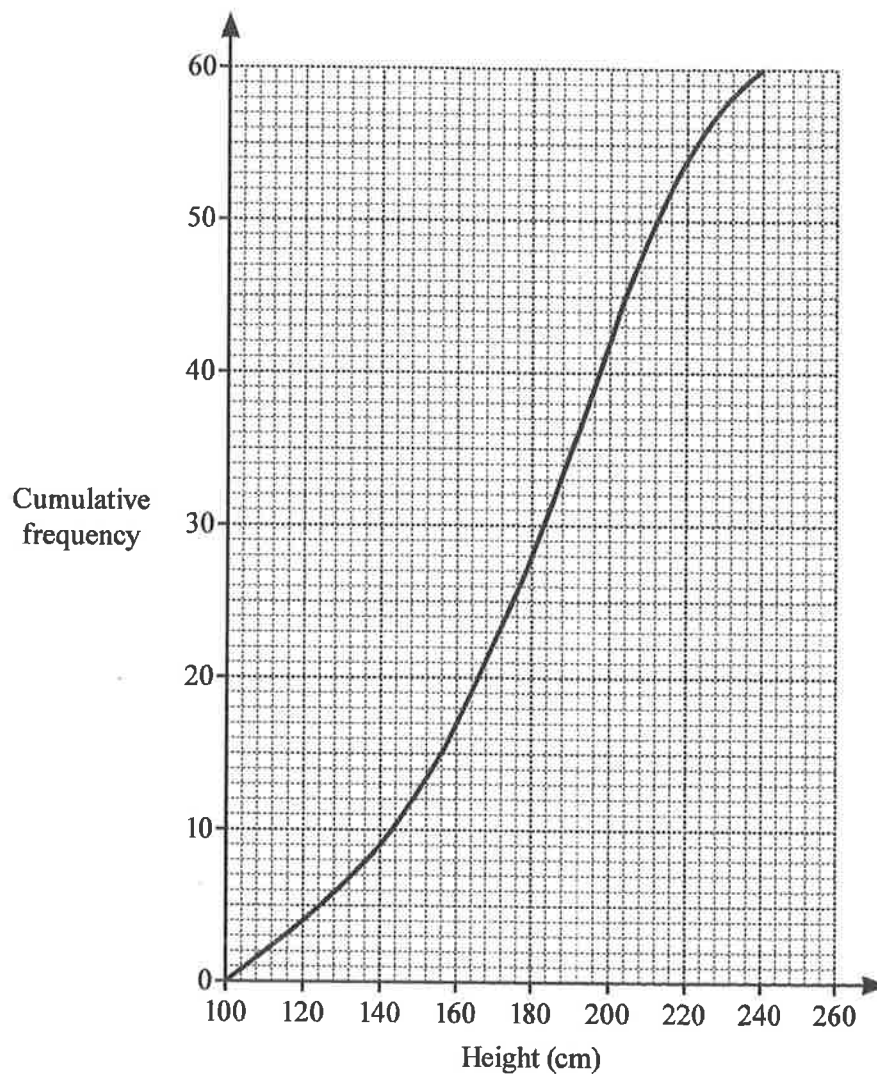
11. (a)



The histogram shows the distribution of the heights of 60 young trees.

- (i) Which interval contains the median height? [1]
- (ii) Calculate an estimate of the mean height of the trees. [1]

(b)



The cumulative frequency graph also represents the heights of the 60 trees.

Use the graph to find

(i) the number of trees with a height of at least 1.92 m,

[1]

(ii) the interquartile range of the heights.

[1]

(c) Another set of trees has heights with an interquartile range of 36 cm.

Explain what this tells us about the second set of trees compared with the first.

[1]

[N19/I/24]

12.

Marriage rates by age group



These charts show the marriage rates for males and females in Singapore for 2004 and 2014.

(a) Make one comment about the ages at which males married in 2014 compared with 2004. [1]

(b) Huma claims that, in 2014, in general males married at a higher age than females.

Do the charts support her claim?

Justify your answer with reference to the charts. [1]

[N18/I/6]

13. The mean height of a group of students is 171 cm.

In the group there are 4 more females than males.

The mean height of the females is 162 cm.

The mean height of the males is 183 cm.

Calculate the total number of students in the group. [4]

[N18/I/15]

14. The masses, in grams, of 30 apples and 25 bananas were recorded.

The results are shown in the stem-and-leaf diagram.

Apples						Bananas					
				5	2	9					
		8	2	1	0	10	3	3	5	5	
	9	6	5	0	0	11	0	2	4	4	7 8
	9	7	4	3	2	12	1	1	5	5	6 8
8	6	5	2	2	0	13	2	3	4	4	9
	9	8	8	6	4	14	0	6	6	x	
			9	9	3	15					

Key (Apples)
2 | 9 means 92 grams

Key (Bananas)
10 | 3 means 103 grams

(a) Write down the median mass of the apples. [1]

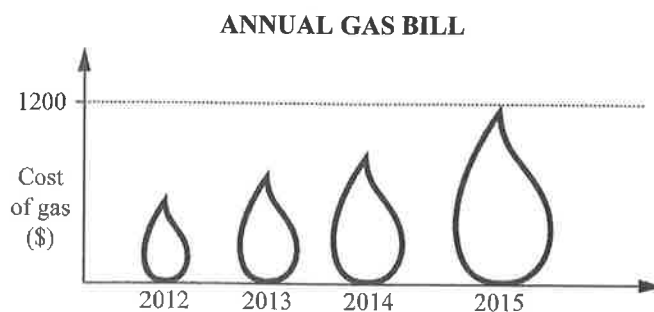
(b) The range of the masses for the bananas is 44 grams.

Find x . [1]

(c) Make two comments comparing the masses of the apples and the bananas. [2]

[N18/I/19]

15. Anil draws this graph to show his annual gas bill for each of the last four years.



State one aspect of the graph that may be misleading and explain how this may lead to a misinterpretation of the graph.

[2]

[N16/I/8]

16. Alec has written down five numbers.

The mean of these numbers is 7, the median is 5 and the mode is 4.

The largest number is three times the smallest number.

Find the five numbers.

[2]

[N15/I/3]

17. Simon weighed six apples.

The mean mass of the apples was 137 grams.

The standard deviation of the masses of the apples was 5.48 grams.

The scales used by Simon were found to be inaccurate.

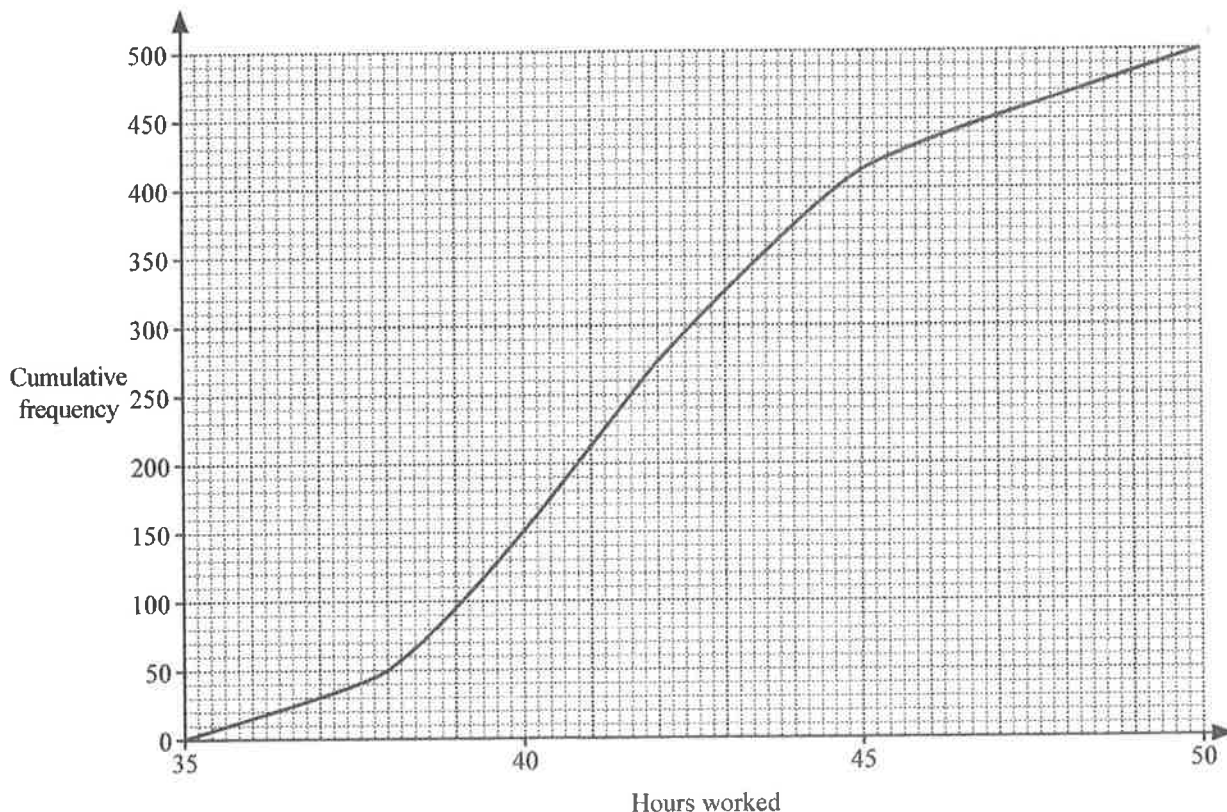
The correct mass of each apple was 25 grams more than Simon recorded.

Write down the values for the mean and standard deviation (SD).

[2]

[N15/I/9]

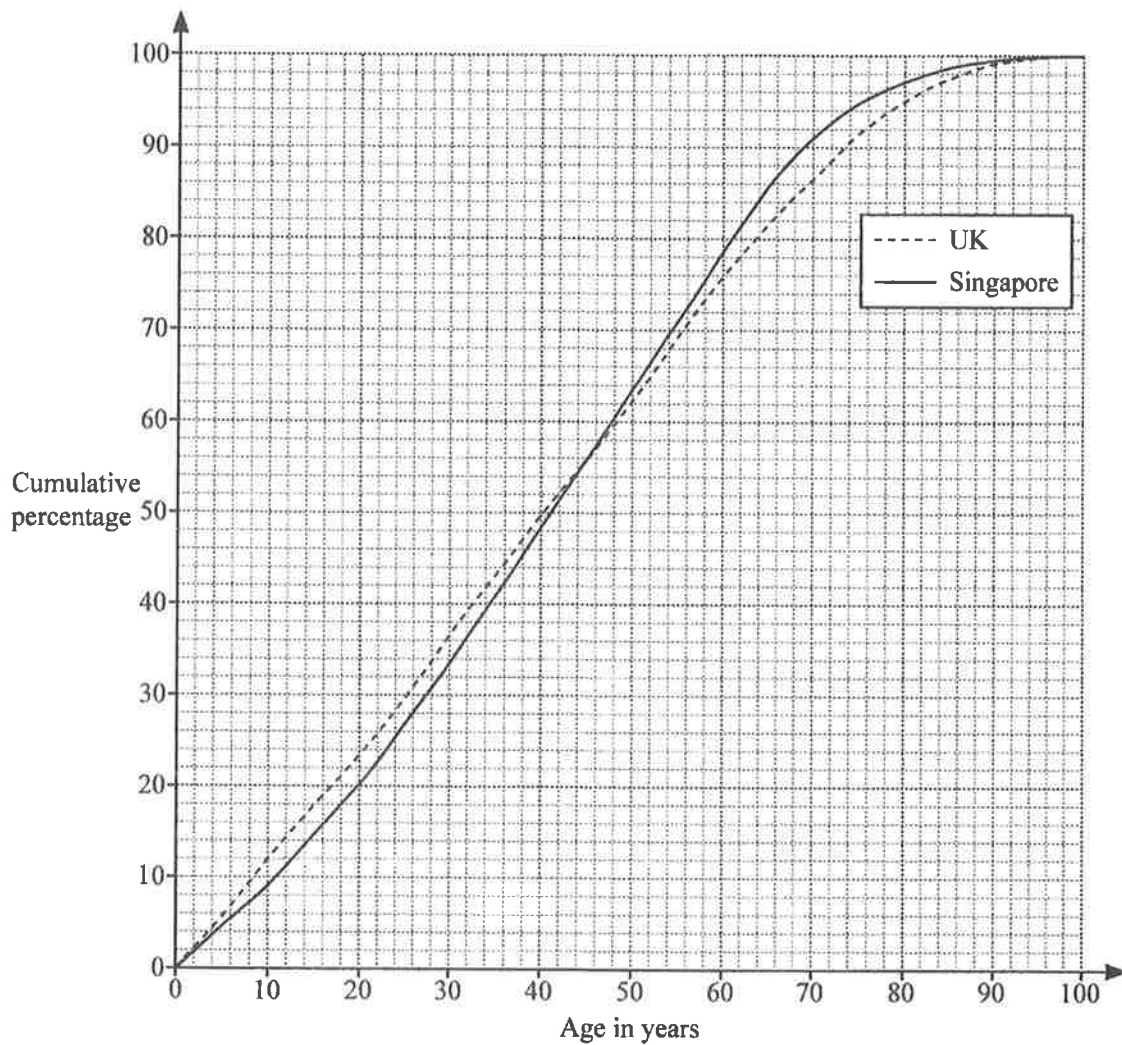
1. A company recorded the number of hours worked in one week by each of its 500 employees.
The cumulative frequency curve shows the distribution of the results.



- (i) Use the curve to estimate
- the median hours worked, [1]
 - the interquartile range of the number of hours worked. [2]
- (ii) In 2017, the average number of paid hours worked in a week in Singapore was 45.1 hours.
Calculate the percentage of the company's employees who worked more than this average. [2]
- (iii) The company wants to reduce the working hours of each of its employees by 3 hours per week.
Describe the effect this change would have on the cumulative frequency curve. [1]

[N21/II/9(a)]

2. The cumulative percentage curve shows the age distributions of the resident populations of Singapore and the UK in 2019.



- (a) Use the curve to estimate
- (i) the median age for Singapore, [1]
 - (ii) the percentage of the UK population aged over 60, [1]
 - (iii) the 80th percentile for Singapore. [1]
- (b) In 2019, the resident population of Singapore was 4.03 million.
Calculate an estimate of the number of people aged under 25 in Singapore in 2019. [2]
- (c) The range of ages for Singapore is the same as the range for the UK.
Make two more comparisons between the age distribution in Singapore and the UK.
Use figures to support your answer. [3]

[N22/II/3]

3. 50 students were each asked how many movies they had watched in the last month.

The results are shown in the table.

Number of movies	0	1	2	3	4	5	6
Frequency	p	10	13	9	6	q	2

The mean number of movies watched by the students was 2.68.

- Find the value of p and the value of q . [3]
- Calculate the standard deviation. [1]
- 50 adults were also asked how many movies they had watched in the last month.

The results are summarised in the table.

Mean	2.04
Standard deviation	1.92

Make two comparisons between the number of movies watched by the adults and by the students.

[N20/II/7]

4. Two groups of students entered a science competition.

Their scores are shown in the stem-and-leaf diagram.

Group A									Group B									
				9	8	3		5		0	1	5	7					
	8	7	6	5	4	2	0		6		2	2	6					
9	7	7	6	5	2	1	1		7		1	3	4	5	5	7	8	9
				8	8	7	2		8		0	6	7					
					5	0			9		1	3						

Key: 3 | 5 | 0 means a score of 53 in group A and a score of 50 in group B

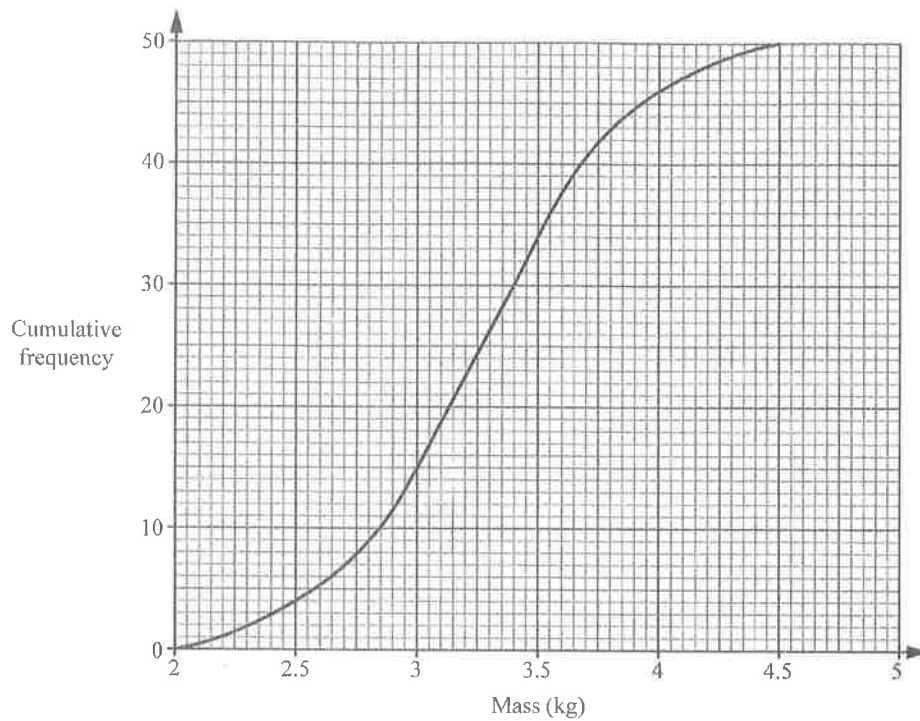
- (i) Find the median score for group A. [1]
- (ii) Find the range of scores for group B. [1]
- (iii) Calculate the percentage of students in group A who scored more than 70. [1]
- (iv) Students who scored more than 70 were awarded a merit.

Students who scored more than 85 were awarded a distinction.

Use this information to make two comments comparing the performance of the students in the two groups. [2]

[N19/II/9(a)]

5. Jiao is researching maternity care in a hospital.
 She searches the internet to find some information.
 She finds information about the masses of 50 newborn babies born in the hospital.
 The cumulative frequency curve shows the distribution of the masses.



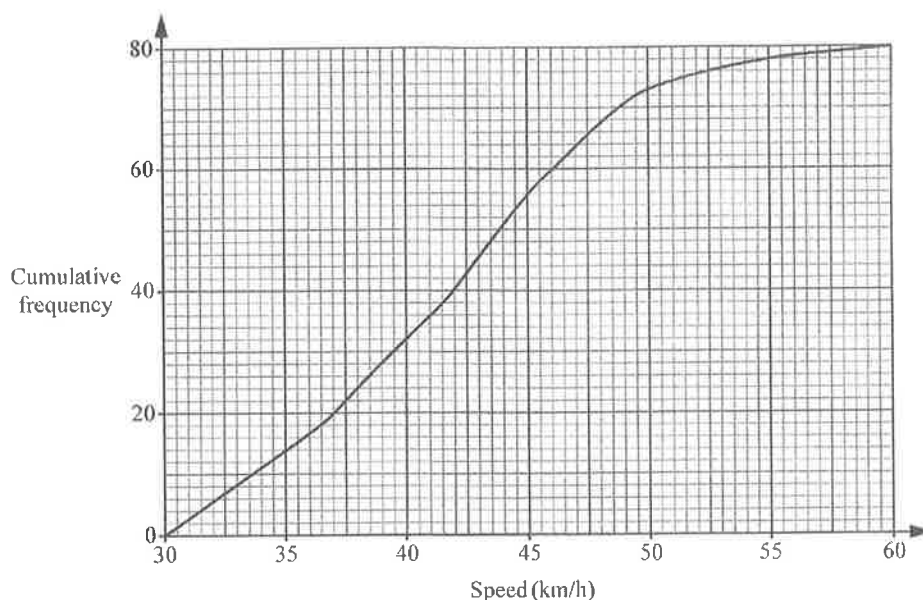
- (i) **Copy** and complete the grouped frequency table for the masses of the babies.

Mass (m kg)	$2.0 \leq m < 2.5$	$2.5 \leq m < 3.0$	$3.0 \leq m < 3.5$	$3.5 \leq m < 4.0$	$4.0 \leq m < 4.5$
Frequency	4	11			

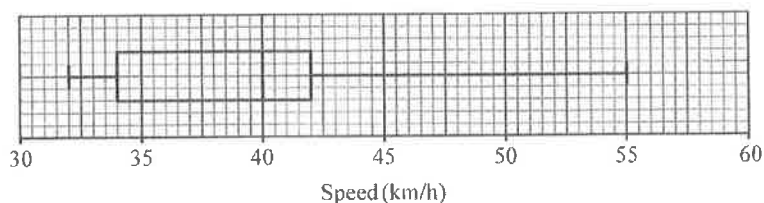
- (ii) Calculate an estimate of the mean mass of each baby. [1]
 (iii) Calculate an estimate of the standard deviation. [1]
 (iv) Explain why the mean and standard deviation are estimates. [1]
 (v) A newspaper article states that 90% of newborn babies have a mass greater than 2.8 kg.
 Comment on whether the data from the hospital supports this claim. [2]

[N18/II/9(a)]

6. The speeds of 80 cars passing a checkpoint one morning were recorded.
The cumulative frequency curve below shows the distribution of the speeds.



- (i) Use the curve to estimate
- (a) the median speed, [1]
 - (b) the interquartile range of the speeds. [2]
- (ii) The speed limit on the road is 50 km/h.
Estimate the percentage of the cars that exceeded the speed limit. [2]
- (iii) The speeds of 80 cars passing the same checkpoint in the afternoon were also recorded.
The box-and-whisker plot shows the distribution of the speeds.



Make two comments comparing the speeds of the cars in the morning and in the afternoon.

[2]
[N17/II/9(a)]

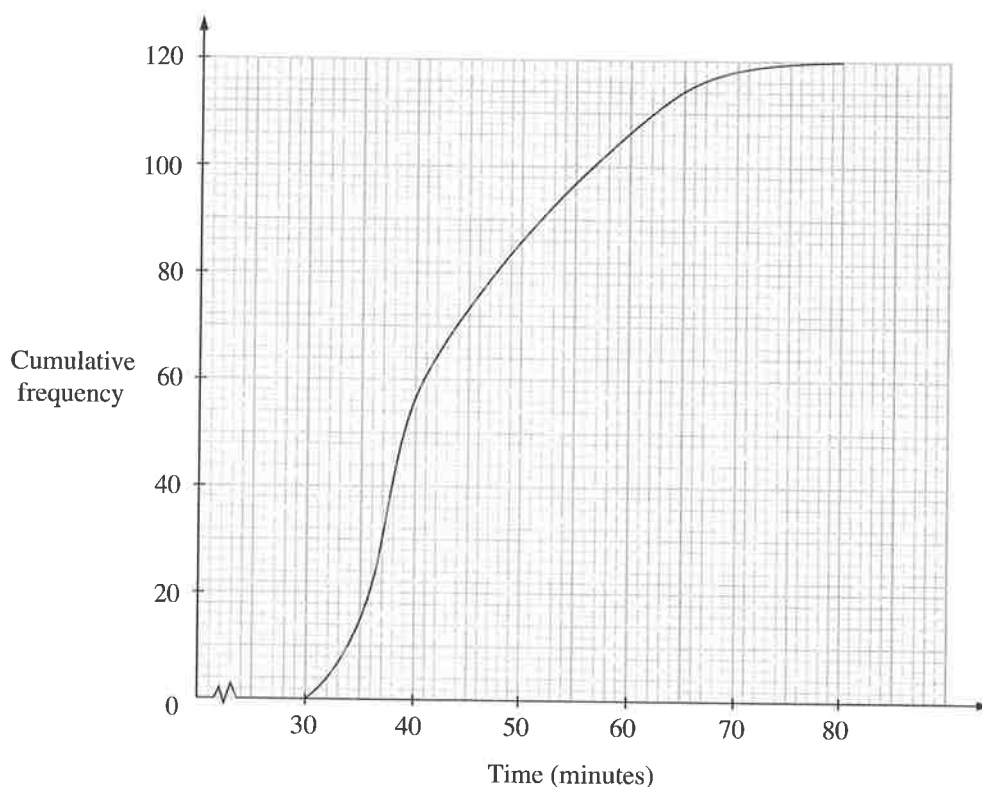
7. A group of 20 students were asked how many minutes they had taken to complete an assignment. The results are shown in the stem-and-leaf diagram.

2	4	5	5	7	7	7	8	8	9
3	1	2	2	2	3	7			
4	4	6	7	8					
5									
6	9								

Key: 3 | 2 means 32 minutes

- The teacher expected the assignment to take 30 minutes.
Find the percentage of students who took longer than expected. [1]
 - Find the median time. [1]
 - Explain why the mean may not be an appropriate average to use to summarise the times taken by these students. [1]
 - Find the standard deviation of the times. [1]
 - The standard deviation of the times taken by a second group of students to complete the same assignment was 8.64 minutes.
Use this information to comment on one difference between the two distributions. [1]
- [N16/II/9(a)]

8. The times taken by 120 men to complete a cycle race are recorded. The cumulative frequency curve below shows the distribution of their times.



- Use the curve to estimate
 - the median time, [1]
 - the interquartile range of the times, [2]
 - the percentage of the men who took at least one hour to complete the race. [2]
 - The times taken by 120 women to complete the cycle race had the same interquartile range as the men's times but a higher median.
Describe how the cumulative frequency curve for the women may differ from the curve for the men. [1]
- [N15/II/10(a)]

